

VortexTech Cyber Security Internship

Week 2 Submission: Hands on with Basic Security Tools

Overview

This submission covers the two Week 2 deliverables. The first is a password strength evaluator written in Python, which scores a password based on length, character variety, common password checks, pattern detection, and entropy. The second is a local port scan performed with Nmap against my own machine (localhost), followed by a short analysis of the results. Scanning was limited strictly to localhost (127.0.0.1), my own device, in line with the task rules.

Part 1: Password Strength Checker

What it does

The checker takes a password and returns a rating (Very Weak, Weak, Medium, Strong, or Very Strong), a numeric score from 0 to 10, an entropy estimates in bits, a pass or fail checklist, and specific feedback on how to improve the password. It also includes a secure password generator.

How the logic works

The evaluation combines several independent checks:

1. **Length.** Passwords earn more points as they get longer. Reaching 8, 12, and 16 characters each raises the score.
2. **Character variety.** The script checks for lowercase, uppercase, digits, and special characters. Each category present adds one point.
3. **Common password check.** The input is compared against a set of extremely common passwords such as 123456, password, and qwerty. A match forces the score to the floor and flags the password as very weak.
4. **Pattern detection.** The script flags repeated characters (aaaa), sequential strings (abcd or 1234), and keyboard walks (qwerty or asdf). Each pattern found reduces the score.
5. **Entropy.** The script estimates entropy in bits from the size of the character pool and the length of the password. Higher entropy adds points.
6. **Breach check.** The password is checked against the Have I Been Pwned database using its k anonymity range API. Only the first 5 characters of the SHA 1 hash ever leave the machine, so the password itself is never transmitted.

Rating scale

Score	Rating
0 to 2	Very Weak
3 to 4	Weak
5 to 6	Medium
7 to 8	Strong
9 to 10	Very Strong

Example outputs

The table below shows real results produced by the script for passwords of varying quality. The breach check was disabled for this table so the values are reproducible offline.

Password	Rating	Score	Entropy (bits)	Checks passed
123456	Very Weak	0 / 10	19.93	2 / 8
Summer2024	Medium	5 / 10	59.54	7 / 8
Tr0ub4dor&3	Strong	7 / 10	72.10	8 / 8
g7#Qm!v2Lp@Xz9Rw	Very Strong	10 / 10	104.87	8 / 8

These results make sense. The all numeric 123456 is a known common password and is correctly rated Very Weak. Summer2024 passes most checks but is fairly predictable, so it lands in the middle. The last example, a long random string with all four character types, reaches the maximum score

The screenshot shows the 'Generate' tab of the ParseItongue Password Security Suite. It features a 'Password length' input set to 16. Below this are four checked checkboxes: 'Uppercase (A-Z)', 'Lowercase (a-z)', 'Digits (0-9)', and 'Symbols (!@#\$)'. There is an unchecked checkbox for 'Exclude ambiguous (!1l1o00l)'. A green 'Generate Password' button is prominently displayed. At the bottom, a 'Generated password' field is shown with a 'Copy' button next to it.

The screenshot shows the 'Check Strength' tab of the ParseItongue Password Security Suite. It prompts the user to 'Enter a password to analyze' with an input field and a blue 'Analyze' button. Below the input field, the text 'Awaiting input...' is displayed, along with a '0/10' score indicator. At the bottom, there are two sections labeled 'Checklist' and 'Recommendations'.

Part 2: Local Port Scan with Nmap

Commands used

I scanned only my own device (localhost), which is always safe. I used a service and version scan (-sV) so that each open port is identified:

nmap -sV localhost

Scan results

Running nmap -sV localhost produced the output below. Nmap reported the host as up, found 7 open TCP ports, and reported the remaining 993 ports as closed.

```
C:\Windows\System32>nmap -sV localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-15 22:56 Arab Standard Time
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00080s latency).
Other addresses for localhost (not scanned): ::1
rDNS record for 127.0.0.1: ic.adobe.io
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
135/tcp    open  msrpc          Microsoft Windows RPC
445/tcp    open  microsoft-ds?
902/tcp    open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
912/tcp    open  vmware-auth    VMware Authentication Daemon 1.0 (Uses VNC, SOAP)
1001/tcp   open  tcpwrapped
1947/tcp   open  http           Aladdin/SafeNet HASP license manager 24.04
5357/tcp   open  http           Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.91 seconds
```

The table below explains each open port and the service behind it:

Port	Service	Version detail	What it is
135/tcp	msrpc	Microsoft Windows RPC	Windows Remote Procedure Call endpoint mapper, used by internal Windows services
445/tcp	microsoft-ds	SMB	Windows file and printer sharing over SMB
902/tcp	ssl/vmware-auth	VMware Authentication Daemon 1.10	Authentication service for VMware virtualization software
912/tcp	vmware-auth	VMware Authentication Daemon 1.0	Second VMware authentication service
1001/tcp	tcpwrapped	not identified	Port answered but closed the connection before Nmap could fingerprint the service
1947/tcp	http	SafeNet HASP license manager 24.04	Local software licensing service installed by a licensed application
5357/tcp	http	Microsoft HTTPAPI httpd 2.0	Windows Network Discovery web service (Function Discovery)

Two observations from this scan:

- Most of the open ports are not part of Windows itself. Ports 902 and 912 were opened by VMware, and port 1947 by a SafeNet HASP license manager. This is a clear illustration of how installing software quietly increases the attack surface of a machine.
- The reverse DNS entry mapping 127.0.0.1 to ic.adobe.io comes from a hosts file entry, not from an open service, so it does not represent something listening on the machine.

Common ports reference

To interpret the scan I looked up the service that commonly runs on each open port. The table below is a quick reference:

Port	Common service
21	FTP (file transfer)
22	SSH (secure remote login)
23	Telnet (insecure remote login)
25	SMTP (email sending)
53	DNS (name resolution)
80	HTTP (web)
110	POP3 (email retrieval)
135	Windows RPC
139	NetBIOS session
143	IMAP (email retrieval)
443	HTTPS (secure web)
445	SMB (Windows file sharing)
3306	MySQL database
3389	RDP (Windows remote desktop)
8080	HTTP alternate

What the results mean

Each open port is a doorway into the machine. An open port is not automatically dangerous, but every listening service is code that could contain a vulnerability, so fewer open ports means a smaller attack surface. In this scan, ports 135 and 445 are standard Windows networking services. Port 445 (SMB) is the most security sensitive of the group, because SMB has been targeted by well known attacks in the past, for example the EternalBlue exploit behind WannaCry, and it should never be exposed directly to the internet. The remaining open ports came from installed software: two VMware authentication services (902 and 912), a SafeNet license manager (1947), and Windows Network Discovery (5357). None of these need to be reachable from outside the machine, so in a real deployment they should be blocked at the firewall for anything beyond

localhost. It is also worth noting that no remote desktop service (RDP, port 3389) or other remote login port was open, which is a good sign.

Reflection: Real World Security Risk

Weak passwords and unnecessary open ports increase a system's risk of attack. Automated tools can quickly crack common or reused passwords, while exposed services may contain vulnerabilities attackers can exploit. Together, these weaknesses make unauthorized access much easier. Using long, unique passwords, enabling multi-factor authentication, closing unused ports, and regularly patching services can significantly reduce this risk.